📊 Population Counters

Name: Class: Date:

Activity 1 · Hands-on · 🌳 Outdoors · ⏱ 1 hour

🟩 Quadrat Count on the Lawn

You will learn: Population density can be estimated by counting individuals in sample squares (quadrats).

YOU NEED

- String (4 m) and 4 sticks to make a 1 m × 1 m square
- Notebook
- Hand lens

⚠️ **Safety:** Watch out for ants and thorns; wear shoes.

STEPS

1. Mark a 1 m × 1 m quadrat on a lawn or open ground.
2. Count the individuals of 3 different plant species inside it (e.g., grass clumps, Tridax, Oxalis).
3. Throw the quadrat at 2 more random spots and count again.
4. Find the mean number per square metre (density) for each species.
5. Which species is most abundant?

OBSERVATION TABLE

Species	Quadrat 1	Quadrat 2	Quadrat 3	Mean density (per m²)

QUESTIONS

1. Why do we use several quadrats instead of one?

2. What is population density?

3. Name two processes that increase population density.

WHAT IS HAPPENING? — THE SCIENCE

Counting every plant is impossible, so ecologists sample. Density = number of individuals per unit area. Several random quadrats give a reliable mean.

Key terms: **population density** **quadrat** **sampling**

Activity 2 · Hands-on · 🏠 At home · ⏱ 30 minutes

🫘 Mark–Recapture with Beans

> **You will learn:** Mark–recapture estimates the size of a population that cannot be counted directly.

YOU NEED

- A jar with a few hundred dried beans (rajma or chana) —• A marker pen
  don't count them!

STEPS

1. Take out a handful (about 30) of beans, mark each with a dot and count them (M).
2. Put them back and shake the jar well.
3. Take out another handful and count the total (C) and how many are marked (R).
4. Estimate the population: N = (M × C) ÷ R.
5. Repeat 3 times, then count all the beans to check your estimate.

OBSERVATION TABLE

Trial	Marked first (M)	Second catch (C)	Marked in catch (R)	Estimate N

QUESTIONS

1. Why must the jar be shaken well?

2. How close was your estimate to the real count?

3. Give one real animal studied by mark–recapture.

WHAT IS HAPPENING? — THE SCIENCE

If marked beans mix evenly, the fraction marked in the second catch equals the fraction marked in the whole population: R/C = M/N. Ecologists use this (with tags or bands) for fish, birds and animals that move around.

Key terms: **mark–recapture** **population size** **estimate**

Digital activity

💻 Population Growth Simulator

> Change r and K and compare exponential (J) and logistic (S) growth.

Do this digital activity in the BAIO app: **https://narendra-bio-sir.github.io/BAIO-Learning-System-By-Narendra-Sir/activities.html#pu2-ch11/2**

> **You will learn:** Exponential growth: dN/dt = rN. Logistic growth: dN/dt = rN(K − N)/K — growth slows as N nears the carrying capacity K.

THINK & WRITE

1. What happens to the logistic curve when you raise K?

2. Which model is more realistic in nature? Why?